



Full-page handwriting recognition and automated essay scoring for in-the-wild essays

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Abstract

Scoring of handwritten essays in school education settings is a time-consuming task. Normalized assessment and prompt feedback enable a student to improve the articulation, comprehension and overall presentation of ideas. In this work, we present a system that can take in input as the images of the essay sheets and outputs the grade/score of the essay. We show a pipelined approach by combining a handwriting recognition model and automated essay scoring. Current handwriting recognition systems show an excellent transcription performance on the existing public domain dataset. These datasets are primarily captured in a constrained manner. The performance and efficacy of these models on unconstrained data are crucial for text understanding. In our work, we adapt an existing full-page handwriting recognition model to the unconstrained handwritten essay dataset. The full page handwriting recognition model is a deep learning model based on CNN and LSTM layers with explicit modules to identify the start of line, line normalization and text line recognition. The unconstrained dataset is from a national essay competition where students upload the essay after scanning the essay. This dataset is wild in nature as the background, margins, text-fonts and the scanning device make it challenging both visually and algorithmically. We have curated a subset of this dataset for all the experiments in this work and intend to make this dataset publicly available. We further analyze the performance on the downstream task of essay scoring using a set of classical handcrafted features and transformer-based contextual embeddings. We have formulated the problem of essay scoring as a regression task. The pre-trained embeddings/handcrafted features, for each essay, are used as representative features for the essay scoring model. Our results show that there is only a slight performance degradation in the essay scoring task due to transcription errors from the handwriting recognition module. We also show analysis with rubric level scores and handcrafted features to develop a subset of features that directly impact the rubric level score on the essay.

Keywords Document analysis · Full-page handwriting recognition · Automated essay scoring

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1 Introduction

Automated scoring of constructed responses such as essays is a key task in school education settings that can enable normalized scoring with timely feedback. It can be used to assess a student's ability to comprehend and present the concepts. A human grader can score the essay with several rubrics under consideration and give a holistic score. Multiple graders sharing the same rubrics for scoring tend to have a good agreement even though the exact score does not match [10]. The manual efforts taken for the scoring are time-consuming and hence might delay feedback. Recent works on automated essay scoring have shown progress to enable scoring done with machine learning approaches which show a reasonable degree of agreement with human grader [3, 9, 13, 25, 26, 29]. Various proprietary and public datasets have been used for performing experiments and documenting results of essay scoring tasks [3, 9, 25, 29]. These datasets are available as essay-text and score pairs.

A significant challenge lies in handwritten essay scoring, where the essay is available as one or more scanned images of the handwritten text [25, 29]. There is no publicly available dataset for the task to experiment with the end-to-end performance. A few attempts have been made with proprietary datasets using the advancements in Handwriting Recognition (HWR) and Automated Essay Scoring (AES) in a pipeline fashion [25, 29, 30]. In this work, we use a self-curated, close-to-real world dataset gathered from a national essay contest organized by a reputed organization for school children. The dataset is wild in that the students have drafted their essays freely without considering any specific format except the essay topic and length. The students wrote the essay on a paper of their own choice with or without page margin as well as background and uploaded the images after scanning with different devices. These variabilities make the in-the-wild dataset challenging to work directly with existing algorithms. We perform handwriting recognition and essay scoring from a subset of this dataset and show the difference it brings to the overall task. We also show the results of individual pipelines on this subset.

The current handwriting recognition technology has various architectures based on CNN, LSTM and temporal convolution based architectures for word, line and page level text recognition [6, 22, 23, 27, 32, 33, 36]. These models need the ground truth transcriptions be available as the locations of the word/line and text transcriptions for word/line level for each handwritten image of the page. The transcriptions are usually done manually at the data preparation stage and are expensive. In our work, we use Start, Follow and Read Model architecture for full page handwriting recognition [34]. This model uses a very less number of images to train the line location detection. We use an existing IAM dataset for the pre-training for line locations while the text recognition model is trained using page level transcriptions of the curated in-the-wild dataset [17]. This saves the transcription efforts needed for line level location marking.

The existing essay scoring works formulate the problem as either a classification or regression task [2, 3, 13, 16, 25, 26, 29, 38]. The various works consider numerous intuitive, classical handcrafted features or deep learning model based representative features. The earlier works use machine learning models like, SVM, Random forest, etc with a combination of handcrafted features [1, 3, 35, 37]. The recent deep learning based architectures with CNN and LSTM layers have also been for extracting the features for the end task of essay scoring along with explicit feature engineering for several intuitive features. There are more recent works which use attention mechanism to enable explain-ability of the model by looking at the text regions explicitly influencing the score [38]. This analysis may

further enhance the capability of the technology to share a meaningful, timely feedback to the students to further improve on their essay.

In summary, the main research questions being investigated in our work are as follows:

- How well does the current state-of-the-art full-page HWR model adapt to an in-the-wild dataset? Could we improve the HWR performance by adding a pre-processing pipeline to target the artifacts of the data due to its unconstrained nature?
- How much is the impact of the transcription errors on the downstream task of automated essay scoring? We perform our experiments with classical handcrafted features and contextually embedded and deep sequence learning-based features for the essays.
- Can a set of low-level features provide feedback to the student to improve their draft?

We further emphasize that the existing handwriting recognition and essay scoring models perform well on the individual tasks with the publicly available datasets. While our work shows the complete pipeline of the experiments with a much challenging in-the-wild dataset. Figure 1 shows an overview of the our system. The input to the system is a handwritten essay page image and the output is the essay score as well as feedback on low-level features. The feedback can help student to understand the actionable areas for improving the score. The contributions of our work are as follows:

- We curate a subset from an in-the-wild dataset for the task of handwritten essay scoring. This dataset can be used for end to end testing for handwritten essay scoring. The data curation involves selecting the text regions and performing manual transcription for each page image for handwriting recognition pipeline. This data can be used for complete pipeline of the overall task, i.e., handwriting recognition followed by essay scoring.
- We show that our curated in-the-wild dataset is much more challenging when used directly with the existing models. For this, we exploit a state-of-the-art page-level handwriting recognition model, a CNN and LSTM based architecture, trained with the publicly available dataset [34]. We adapted the pre-trained full-page handwriting recognition model to our in-the-wild dataset. The training paradigm that we have used requires only page level transcriptions for each page image of the in-the-wild dataset and does not require intensive human efforts for marking the line location information for each line of handwritten text.
- We show that the full page handwriting recognition does not get better by adding a pre-processing on the dataset to remove some artifacts like watermark and back imprints. However, the quality of the image after the pre-processing appears to be better for text recognition.

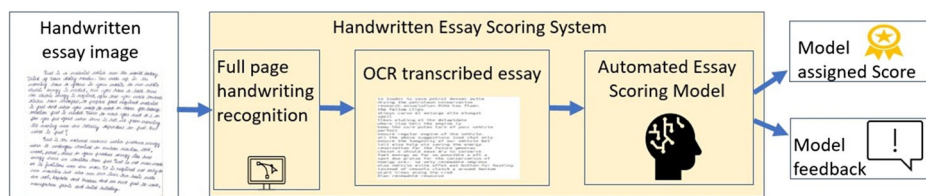


Fig. 1 Overview of our Handwritten essay scoring system. The system considers a full page handwriting recognition system followed by an automated essay scoring system. The essay scoring system also enables feedback mechanism that can give insights to the student on the improvement areas

- We show that the current handwriting recognition pipelines have high errors, but the automated essay scoring task does not significantly impact. We show our results with both classical handcrafted features and pre-trained BERT-based embeddings.
- We show a rubric level analysis for the essays, and the set of features from the handcrafted features that directly impact these scores. This analysis can help share a feature-level feedback with the students.

Our paper is organized as follows. In the following section, we review related works. Section 3 presents details of a much challenging in-the-wild dataset of handwritten essays used in this work. Sections 4 and 6 details our approach for full-page handwriting recognition and essay scoring, respectively. Section 5 demonstrates an experiment to pre-process the HEW dataset to reduce the water mark and back imprints on the essay sheets. These artifacts are present in some of the essay sheets only. The effect of such pre-processing and considerations are present in detail. Section 7 presents a rubric level analysis of the essay scoring task to come up with a set of features impacting a given rubric. Section 8 gives an analysis and discussion around the complete pipeline. Finally, Section 9 concludes the paper with future directions.

2 Related works

Many approaches have been proposed for handwritten short answer assessments. An early work by Srihari et al. considered the scoring of short handwritten essays [30]. These essays were written in response to the question asked on a given reading comprehension on “American First Ladies”. The approach involved a word-level segmentation and recognition of the essay text followed by Latent Semantic Analysis (LSA) for essay scoring. This earlier work showed possibility of using handwriting recognition and automated essay grading in a pipeline manner for the end-to-end task of handwritten essay grading. However, it needed human intervention at multiple stages with closed vocabulary approach for handwriting recognition. The essay grading module based on LSA features showed a low mean average grade but lacks the explainability for the semantic and syntactic understanding of the text passage.

More recent work on essay scoring by Rowtula et al. formulates the task as a self-supervised classification task [24]. The features extracted are fine-tuned for each question without any direct supervision. The work proposed semantic analysis for essay scoring with derived features in handwritten text space using information retrieval and extraction and Natural Language Processing(NLP). This method, being self supervised, requires less efforts for ground truth scores for the essays, yet needs a textual reference answer for keyword spotting. The work focusses on scoring based on contextual similarity of the recognized words and lacks the understanding on semantic, syntactic and coherent structure of the essay.

Sharma et al. showed the efficacy of line-level handwriting recognition followed by essay scoring model [25]. They also presented an analysis of static embeddings and various model configurations for essay scoring [26]. This work requires location information for each line in page image which requires intensive efforts for the data preparation for handwriting recognition task. This method requires additional line recognition module. The various model configurations show the efficacy of pre-trained and fine tuned embeddings of words for the task of essay scoring.

A natural approach for the overall task is to consider transcribing the handwritten text on a word or line level followed by exploiting the text essay scoring system [25, 26, 29, 30]. This approach benefits from the recent advancements in both handwriting recognition and essay scoring systems. In the following subsections, we summarize the related works in both HWR and AES.

2.1 Handwriting Recognition (HWR)

HWR, being an image-based sequence modelling task, is approached using Recurrent Neural Networks (RNN) and Connectionist Temporal Classification(CTC) as an imperative structure in various combinations to perform the word or line-level recognition of characters [7, 22, 23, 32]. CTC cost function aids in more straightforward ground truth text and does not need ambiguous label, i.e., character boundaries [8]. A combination of recent data augmentation and regularization methods have been shown to improve the performance of these models [22, 23, 33]. Some recent works have further optimized number of parameters, training, and inference time [4, 23, 27]. All these models require segmented word or line-level images with corresponding manual transcriptions for the model training. As mentioned earlier, these models requires annotations on word or line level and hence are effort intensive for preparing each dataset. Therefore, we concentrated on full page handwriting recognition for the page images of essays in our dataset. This approach requires the page level transcription for each page image from the training and validation dataset.

More recent works have proposed models for full-page handwriting recognition. Moysset et al. proposed a regression network for text line detection and localization [21]. They further extended the experiments to show the results for full-page text recognition in heterogeneous documents. However, the network does not work well with extended text line images with multiple size text in a line. Yousef et al. proposed Origaminet, which can augment any CTC trained text line recognition model to full-page handwriting recognition model [36]. The model does not need any explicit visual and textual localization information, and the training data is composed of unsegmented image and text pairs. The work does not address the problem of multi-column documents directly and also is limited with the maximum number of characters on single page image. Wigington et al. proposed to use three different networks to find the start of line position, line segmentation, and text recognition [34]. The work, formally called Start Follow and Read (SFR), takes a small sample of images with the start of line and line segmentation annotation for pre-training the networks. All the pre-trained networks are then combined and trained in an end-to-end fashion with only full page images and corresponding text transcriptions with line breaks. The network can work with multiple formats, including the multi-column, multi-scale, warped text regions. This model shows good performance on text line segmentation and transcription on the images without any explicit background.

2.2 Automated Essay Scoring (AES)

Earlier essay scoring models concentrated on count-based handcrafted features that captured the length, vocabulary, and grammar features [1, 3, 37]. A broad set of count and occurrence features include count or ratio of nouns, Token n-gram, POS n-gram, conjunctions, adjectives, adverbs, and verbs in the essay text [3, 37]. Some high-level features include a measure of readability, syntax, cohesion, pairwise sentence similarity, and adherence to the prompt [3, 35]. These features enable a strong feature set but still lacks the proof of language understanding and can be attacked with adversarial samples with easy efforts. Most of

these approaches use a combination of these features to train a supervised machine learning model for in-domain essay scoring. The ground truth in most of the dataset is a combined score from multiple graders. The earlier works show the essay scoring results with different metrics on various proprietary dataset [24, 29, 37, 38]. ASAP-AES dataset¹ is a widely used publicly available dataset for showing the experimental results of automated essay scoring tasks for comparison of approaches.

Some recent approaches use pre-trained dynamic contextual or static embeddings for representing words or phrases in the essays and use these features for the regression or classification task of essay scoring [16, 25, 26]. These pre-trained embeddings are trained on vast corpus and learn a better representation of words [2, 5, 11, 12, 14, 18–20, 31]. Contextual embeddings derive the representation of the word/phrase by considering the context it is appearing in. These embeddings are widely used in various NLP downstream tasks and show promising results even though not fine-tuned on task-specific dataset [2, 12, 31].

In this work, we use an unconstrained dataset of handwritten essays by school students. While existing works used publicly available dataset which have some constraints defined for both the HWR and AES tasks, we use an in-the-wild dataset which is unconstrained in nature. The dataset description is given in the next section. As opposed to Origaminet, the SFR model allows more variability in terms of article length and does not restrict the number of handwritten characters in the input page image [36]. We, therefore, adapt the SFR model to our dataset for the full-page handwriting recognition task [34]. This model is an end-to-end trainable model which can segment the text line images which are not necessarily parallel. This setting, in particular, is suitable for our dataset as we have some lines of text that are not necessarily parallel. Further, we use pre-trained contextual embeddings with the LSTM regression model for essay scoring tasks. The pre-trained embeddings enable enriched embedded representation by considering the context in which each word in the essay is appearing. We present the results of essay scoring with both manually transcribed and OCR-transcribed essays. The details are presented in Sections 4 and 6.

3 Dataset

In this section, we describe our unconstrained or in-the-wild handwritten essay dataset. The dataset was collected as part of a national-level essay writing competition. The participants are school students of grades 7, 8, 9 and 10. They were asked to submit images of sheets of handwritten essays with word length 700. The competition guidelines specify the topic of the essay–“Save Fuel for Better Environment”. The competition accepts entries in multiple languages. Our work considers essays written in English only. We will refer to this dataset as the Handwritten Essays in the Wild (HEW) dataset.

The dataset is available as images of handwritten essays and associated scores assigned by human graders. These images are either scanned from a flatbed scanner or mobile camera. There is high variability in the dataset concerning lighting, size of the sheet used for writing, cropping region, background, illustrations, text-size, text-highlights etc. The total

¹<https://www.kaggle.com/c/asap-aes/>

score assigned by human graders is the addition of the score at four different rubrics. These rubric level scores are also available in the dataset. Following are the rubrics:

- Focus, Clarity, and Details on the topic
- Importance of Fuel Conservation for India
- Critical Thinking, Innovative Ideas Suggestions
- Presentation, Cohesiveness, Language, Grammar

In this work, we examine the suitability of existing methods for automated scoring of handwritten essays on this unconstrained dataset. For this reason, we kept the pre-processing to a bare minimum. We are considering only text essays in this work, did not use essays containing any illustrations in this work. We cropped the images to select the text areas only. Considering the essay scoring scenario where additional information can introduce bias and the privacy concerns, we also cropped out the student's information wherever present.

The full dataset consists of more than 9000 essays. The shortest and the longest essay in the training dataset has 66 and 1236 words, respectively. We sampled a subset of 355 essays and performed the manual pre-processing specified above. We did not specifically consider sampling essays based on scoring or handwriting style. Some of the example essay images are shown in Fig. 2 from different essays. These samples show some variabilities that our dataset possess. Figure 2c–d show pages with explicit margins which to some extent indicate start of line position. Figure 2a and d show the examples where some text lines are with different orientations of the baseline on the same sheet i.e., not all lines are parallel. Figure 2b–e show the examples where some background is appearing in the form of ruled line sheets or visible background impressions from two-sided used sheets.

We further divide our dataset in subsets of 22 and 333 essays for handwriting recognition and essay scoring pipeline experiments respectively. This was done to keep the handwriting for HWR and AES pipeline disjoint. The relatively small subset of handwriting recognition is taken as it is used to fine-tune an existing handwriting recognition model, while for essay scoring, we show model training results from scratch as well as fine-tuning. We use a full-page handwriting recognition model, which does not require us to crop the lines in a pre-processing pipeline. Further details are presented in the following section.

We also use the IAM dataset to pre-train the handwriting recognition model and ASAP-AES dataset for the pre-training essay scoring model [17].

4 Full-page handwriting recognition

Our handwriting recognition approach is based on an existing method by Wigington et al., that takes as input a full page of handwritten image (containing multiple text lines) and outputs the transcriptions along with the position and segmentation of line images [34]. The following subsection gives a brief summary of the work formally called “Start, Follow and Read” or SFR and its adaptation for the HEW dataset. Section 4.2 shows the transcription performance of this model on the HEW dataset.

4.1 Start, follow and read model

The SFR Model has the following task-specific modules:

- Start of Line (SOL) finder network, a Region Proposal Network (RPN). It gives out the possible start of line locations, the scale of the characters in the line image and

the direction of next characters in line. The architecture is based on a truncated-VGG network.

- Line Follower (LF) Network, which segments and normalizes each line image. The network architecture is a 7 layer CNN with fully connected layer to regress the direction and height of the text line in incremental steps. Beginning from the start of the line, this model follows each handwritten line iteratively and can follow curved lines in our dataset. It returns a normalized line image [34].
- Text line recognition Network is also the same CNN architecture as LF network followed by Bidirectional LSTM and CTC layers. It takes in the normalized line image as input and performs a line-level handwriting recognition task.

We have used the same SOL and LF network architecture as used by [34]. For HWR Model, we have changed the input size of the images to $w \times 32$ as opposed to $w \times 60$ in the original paper. This change was done to match the computational constraints of our available hardware resources. Each of these modules is first trained separately with a small amount of data and then in an end-to-end manner on the entire dataset. Wigington et al. showed quite promising results on RIMES (French) dataset and IAM (English) full-page dataset [34]. The work uses multi-line handwritten text from the IAM dataset.

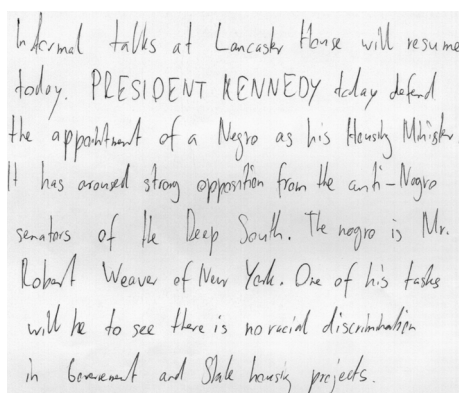
4.2 SFR for HEW dataset

An example image from the IAM dataset is shown in Fig. 3. The inter-line separation between the consecutive lines is good. Also all the images have the same plain background. These images appear to be taken in much constrained way as compared to HEW dataset as shown in Fig. 2.

Under the settings given in the original work, we used the pre-trained model, trained on the IAM dataset for page-level handwriting recognition, for the HEW dataset. The SOL and LF network results are overlaid on a page image from the HEW dataset and shown in Fig. 4a. In our experiments, the SOL network fails to detect most of the start of line positions in our data, mostly when the left margin line is present. For some lines on a ruled sheet, it detects multiple start of line for a single text line. The LF network also misses to follow some consecutive lines properly when the lines are not parallel.

To overcome these issues, due to the unconstrained nature of the HEW dataset, we perform a fine-tuning of the SFR model (All three modules in an end-to-end training) which was pre-trained using only IAM dataset. We use about 22 manually transcribed essays from

Fig. 3 A sample page image from the IAM dataset



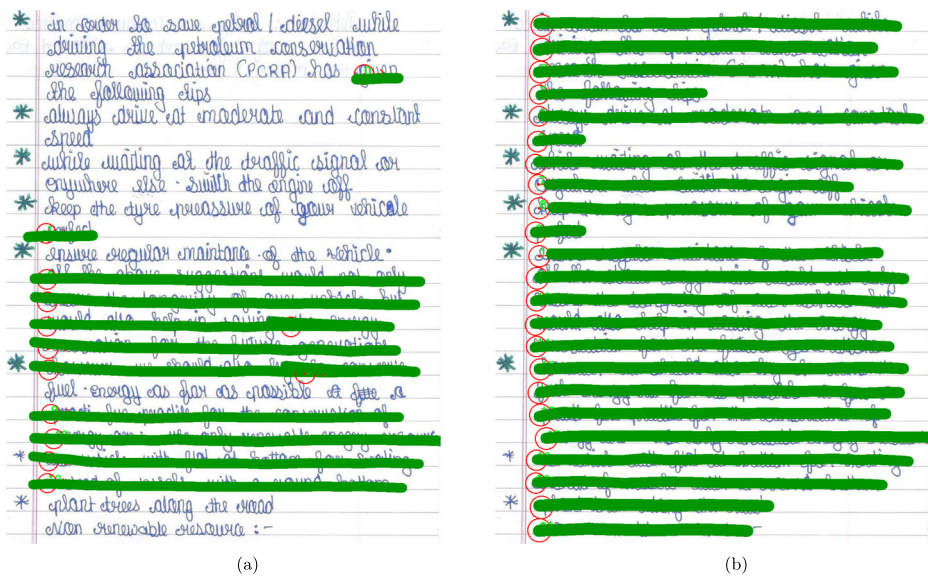


Fig. 4 Output of SOL(The center and radius of the red circle shows start of line and scale of the text in the line.) and LF (Green fill showing the segmented line image) network overlaid on the page image. The results obtained from SFR model trained using the IAM dataset and finetuned on the HEW dataset are shown in 4a and 4b, respectively

the HEW dataset spanning over 131 full-page images/sheets. Figure 4b shows the output of fine-tuned SOL and LF networks overlaid on the same handwritten page image as Fig. 4a. This page image is from a test set which is not a part of the training set for fine-tuning.

4.3 End-to-end performance of the SFR model

The test set of end-to-end transcriptions is a collection of 333 essays spanning over 1084 images of the essay text. None of the writers for essays in the test set are the same for the essays in the train set. We compared the transcription results of the OCR transcribed essays with the transcriptions done by a human. The transcription performance of the model is measured in terms of character and word error rate computed over the essay level transcriptions. The Character Error Rate (CER) is the mean of the normalized edit distance of the OCR-transcriptions and human assigned transcriptions, computed over all essays. The Word Error Rate (WER) is the same as CER with edit distance calculated over word level. The CER and WER for the test set are 0.21 and 0.51, respectively. Figure 5a shows the raw transcriptions received from fine-tuned SFR model.

We use a 3-gram character level Language Model (LM) decoding to bring transcriptions close to natural language. We use kenLM for training LM with modified Kneser key smoothing. The corresponding CER and WER for the manual transcriptions are 0.15 and 0.32, respectively. The transcription results after decoding with the language model are shown in Fig. 5b. In the next section, we will use these transcriptions with LM decoding to perform the downstream essay scoring task.

Fuel is a material wthich run the world today
 think of Hour daily routine. Hou llake ep in the
 morning have a eglance in lyour mobill to un mobile
 electril enegl is niedld then you have a lath thire
 loo clutric energy is riquired after that fou walk toward
 kitihen have bilatfast, to peepare food required material
 is fuel and iuhenn yout tualk to work or learn for biansp
 ovation fuel is nceded. When to reach you need of l or
 fan you fecl good luhenn there ts hot lo from morning
 till evening wer care totally dependent on fuel. But
 siihat is fuel
 Fulee is the natural resource which produces energy
 iithen it undergoes thetical or pullear reation, loal
 lrpod petrot diesel or gases produlce energy like hitat
 energy shenle aire tonsides them fuel Fuel is not man made
 so its fudicious use are more. It is required hot only to
 rueln mauhines buud also run our livers our liasic needs
 are roli, kapada and Mhatean And we need fuel to cook
 inaneyacture faterie and build buildings.

(a)

Fuel is a material which run the world today.
 think of hour daily routine. You lake up in the
 morning have a glance in your mobile to in mobile
 electric energy is needed then you have a lath there
 too clutching is required after that you walk toward
 kitchen have blatant, to prepare food required material
 is fuel and when bout talk to work or learn for beans
 ovation fuel is needed. When to reach you need of l or
 fan you feel good then there is hot to from morning
 till evening were care totally dependent on fuel. But
 what is fuel
 Fuel is the natural resource which produce energy
 then it undergoes chemical or nuclear reaction, coal
 rod petrol diesel or gases produce energy like hiat
 energy hence are consider them fuel Fuel is not manmade
 so its judicious use are more. It is required hot only to
 ruin machines but also run our lives our last need
 are role, capacity And we need fuel to cook
 manufacture fabric and build buildings

(b)

Fig. 5 Transcription Results for example image shown in Fig. 2a from HEW dataset. Figure 5a shows the results without language model and Fig. 5b shows the results with a 3-gram character language model

5 Pre-processing to enhance the image quality

As mention earlier, the HEW data set has a wide variety of images with different artifacts that might impact the quality of transcriptions received from the handwriting recognition module. In this section, we show the pre-processing steps to enhance the document images by illumination correction and background/watermark removal in Sections 5.2 and 5.1, respectively. The enhanced images are used to train and evaluate the SFR model for

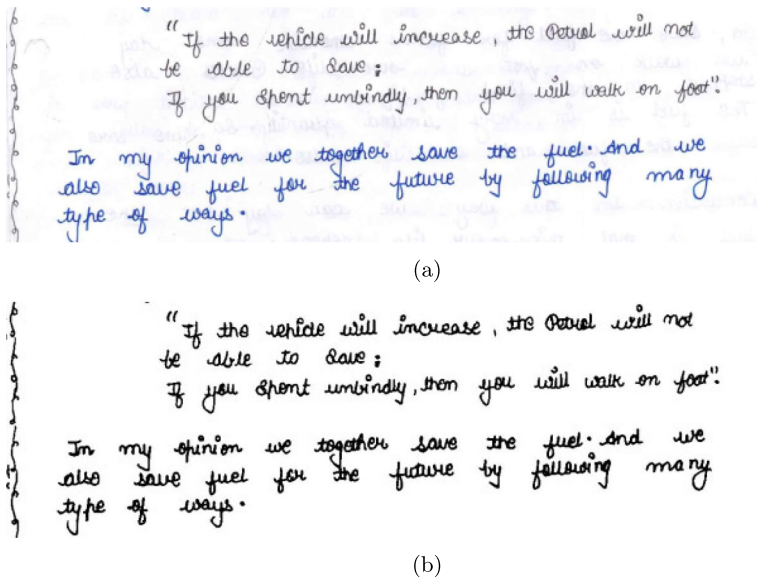


Fig. 6 Removal of back-imprints in documents from DE-GAN watermark removal pipeline. Figure 6a shows the input image with back-imprints and 6b shows the back-imprint removed image from DE-GAN

full-page handwriting recognition. Section 5.3 shows both qualitative and quantitative results of the pre-processing pipeline added in this section.

5.1 Enhancement using DE-GAN

Souibgui et al. proposed DE-GAN for enhancing the degraded documents [28]. The DE-GAN architecture is a conditional GAN which has a U-Net inspired generator and a fully convolution network architecture as discriminator. The authors use DIBCO 2013, DIBCO 2017 and H-DIBCO 2018 datasets for training and evaluating the proposed architecture for the document enhancement. The training is done with paired examples of degraded document images and corresponding clear document images. The network uses a combination of log-loss between the input paired examples and discriminator loss for training. The log-loss in the network helps to keep the document content preserved.

The authors release the pre-trained models² under different settings considered in their work. We use pre-trained model weights released by authors and use the watermark removal pipeline for enhancing the document images in HEW dataset. Figure 6 shows the input and output images from DE-GAN watermark removal pipeline. The back-imprints in the example image in Fig. 6a are removed quite well by the DE-GAN watermark removal.

Figure 7 shows an example image from HEW dataset which contains watermark as well as visible uneven shading effect in the document due to lightning conditions. Images with different shading regions are much complex to clean and can result in higher translation errors when fed directly to handwriting recognition systems. Figure 7b shows the image after watermark removal with DE-GAN. This pre-processing step removed the watermark quite effectively without affecting the text areas where watermark was present in original

²<https://github.com/dali92002/DE-GAN>

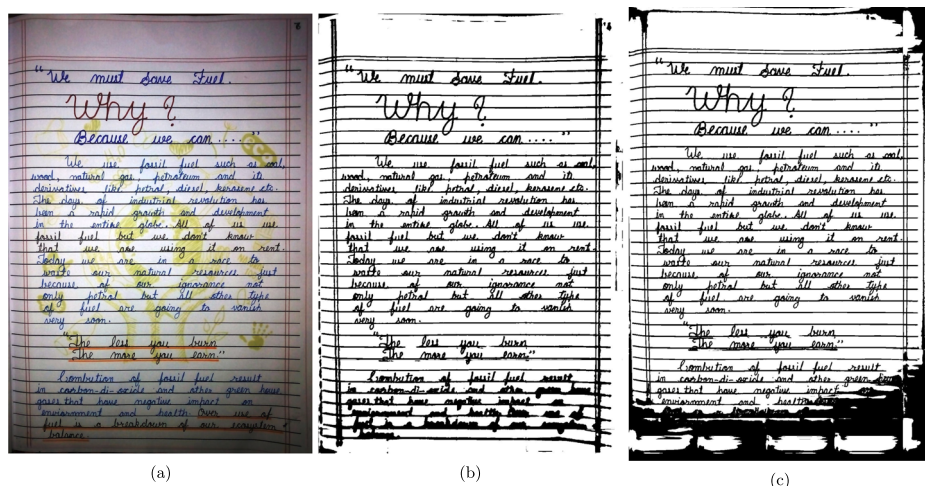


Fig. 7 The qualitative results from pre-processing pipeline. Figure 7a An example image with watermarks from HEW dataset. Figure 7b Watermark removal from DE-GAN. Figure 7c Illumination correction followed by watermark removal

image. However, the text areas in the bottom part of the image, with shadings because of lighting conditions, get affected severely that the document looks degraded qualitatively to a human observer. In order to get a better document text area as well as water removal, we add an illumination correction step before watermark removal. The details of this illumination correction are shown in the following subsection.

5.2 Illumination correction

In order to deal with shading effect, we use the illumination correction network as used by Li et al. [15] for document rectification and illumination correction. The architecture of illumination correction network contains 3 CNN layers followed by residual blocks. The model is optimized with a combination of pixel-wise L1 loss and perceptual loss. We use the pre-trained model³ weights from authors of the paper and get illumination corrected images for all the images in HEW dataset. For the entire pre-processing pipeline, we used this illumination correction step followed by watermark removal. Figure 7c shows the results of document enhancement with the entire pipeline. There are some degradations due to shading effects in the bottom of the document. As compared to the Fig. 7b, the text regions in the bottom area look much readable qualitatively. We therefore use the entire pre-processing pipeline for training the SFR model. The details of the results for handwriting recognition are shown in next subsection.

5.3 Performance of SFR model on quality enhanced images

The pre-processed images, using the pipeline mentioned in this section, from HEW dataset are used for training the SFR model with similar settings as mention in Section 4. We evaluated the transcriptions of the test set images with and without LM decoding. With

³<https://github.com/xiaoyu258/DocProj>

LM decoding, the average CER and WER are 0.21 and 0.39, respectively. Although the pre-processing pipeline shows qualitative improvements in image, the overall quantitative results on the test set does not improve. This experiment shows the effectiveness of the SFR model to work on raw document images as well as the complexity of the document images in HEW dataset.

6 Essay scoring

For essay scoring, we have considered two different models - A combination of handcrafted features with random forest classifier and a transformer-based contextual embedding with LSTM regression model. Both the models were chosen by performing several experiments on the ASAP-AES dataset. We further show the model performance for essay scoring on the HEW dataset. We divide the dataset to contain 200 essays in the training set and the remaining 133 essays in the test set. We train the model using only manually transcribed essays. This decision is taken to avoid introducing any noise in the essay scoring pipeline which might occur during OCR transcriptions. We consider two pipelines for testing - for manually transcribed essays; for OCR-transcribed essays to account for transcription errors due to the OCR pipeline. The following subsection shows the common concerns for both the models, followed by the description of both the models.

6.1 Metrics and score normalization

The range of the scores is different for each prompt in ASAP-AES and HEW dataset. We used score normalization to bring the score to a range (0-1). For each prompt, we used the following for score normalization:

$$normalized_score = \frac{Score\ assigned\ by\ manualgrader}{max_score - min_score + 1} \quad (1)$$

Where max_score and min_score are the maximum and minimum of the score range specified for the prompt. For the HEW dataset, the holistic score ranges from 0-50. The rubric level score range for the rubrics “Focus, Clarity, and Details on the topic” and “Critical Thinking, Innovative Ideas Suggestions” is 0-15. Further, for the rubrics “Importance of Fuel Conservation for India” and “Presentation, Cohesiveness, Language, Grammar”, the score range is 0-10.

We used Mean Square Error (MSE) as a loss function during training and Quadratic Weighted Kappa (QWK) as a metric of performance of the essay scoring task [37].

6.2 Essay scoring with handcrafted features

This section uses several handcrafted features used widely in essay scoring literature to model various essay scoring aspects. We have organized these manually handcrafted features into different categories as follows:

Count based features These are features that deal with the count of the characters and words in the essay. These features include word length, word count, character count, unique word count, and spelling error count. It also includes the count of each of the POS tags like the noun count, verb count, etc. These are simple and effective features. These features

alone are easily attacked by adversarial samples. Therefore, these count-based features are mostly used in conjunction with other complex features.

Sentence based features The number of words in a sentence can measure the degree of complexity of the essay. We further added features to the set where it detects if the number of words in a sentence is in the range (0-10), (11-18), or (18-25). We also use the mean and the variance of the number of words in a sentence per essay.

Readability features Different levels of readability can be associated with different ranges of the score for the essays. This feature implies the preference of writing style. We add the readability metrics like Flesch, Dale Chall, Gunning Fog for our features for essays [37].

Syntactic feature Chen et al. use the mean height and the maximum depth of the parse tree, for each of the sentences in the essay, as the syntactic feature of the essay [3]. We also complemented this using the formality rate, which considers the frequencies of different word classes [9, 37]. We also use bigram and trigram of POS n-grams [3].

Prompt relevance features For prompt adherence, we use the TF-IDF similarity and the semantic similarity using the GloVe embeddings between each of the essays and the prompt [3, 37]. We also use word n-gram features for the most frequent bigrams and trigrams in the essay set [3].

Error features Punctuation, grammar, or spelling errors are some of the most prominent measures of the quality of the essays. We use language tool⁴ which flags simple rule-based errors.

Coherence features The overall coherence of the essay is measured by taking the mean of all sentence pairwise similarity score [35]. The hypothesis is that the degree of semantic relatedness between adjoining sentences serves as a proxy for local discourse coherence.

We use all these features with a random forest classifier in a supervised learning setting where scores are normalized as mentioned in Section 6.1. The results of 10 runs are averaged and presented in Table 1. The QWK score for manually transcribed test set essays is 0.58 and that with OCR transcribed test set essays is 0.56. This low value of the agreement for the OCR transcribed test set with the ground truth is justified as our OCR pipeline has errors associated with it, resulting in spelling or grammar errors.

6.3 Essay scoring with contextual Embeddings

In this section, we describe our essay scoring model, which is a simple 2-layer LSTM regression model. The features are extracted by dividing the essay into contiguous chunks of 64 words and then representing each chunk by contextual embedding from a pre-trained model. We have used BERT as contextual embedding. The use of contextual embedding enables the representation to be influenced by word usage and the context in which each word is appearing. The LSTM layers makes use of sequential features and context for the essay texts.

⁴<https://dev.languagetool.org/>

Table 1 The results of essay scoring model

Features and Model	Test set	QWK	
		Model Trained from scratch	Fine-tuned Model
Handcrafted Features and Random Forest	Manually transcribed	0.58	0.62**
	OCR-Transcribed	0.56	0.47**
Contextual Embedding and LSTM	Manually transcribed	0.58	0.61
	OCR-Transcribed	0.54 0.57	

The performance is shown with an average QWK, of 10 runs, as the metric for both manually transcribed and OCR-transcribed essays from the HEW test set. The results marked with ** are computed by training the model by combining essays from the ASAP-AES training set and HEW training set

6.3.1 Approach

As an initial step, we train our 2-layer LSTM regression model using the ASAP-AES dataset. We use a batch size of 128, 16 hidden units in each LSTM layer, and a starting learning rate of 0.001 with an ADAM optimizer. The model training is stopped after 100 epochs or no improvement in validation loss after 20 epochs, whichever occurs first. The average QWK on the ASAP-AES test set for all the prompts is 0.71.

6.3.2 Essay scoring for HEW dataset

The HEW dataset has only 333 essays with transcriptions. We trained the model with HEW dataset, described in Section 6.3.1, from scratch, and the corresponding results are presented in Table 1. The average QWK for manually transcribed and OCR transcribed essays is 0.58 and 0.54, respectively.

The number of essays is less in the HEW dataset than that for each prompt in the ASAP-AES dataset. In the second approach, we exploit ASAP-AES dataset for pre-training the same model structure and fine-tuned it with the manually transcribed essays in the training set of HEW dataset. The corresponding results are also presented in Table 1. The agreement with the human grader improves for both the manually transcribed and OCR transcribed essays.

The lower QWK for OCR transcribed essays than the manually transcribed essays is justified as the OCR-transcribed essays have some errors due to imperfections in the OCR pipeline. This observation is true for both the training scenarios. We further believe that increase in the number of essays for training this model can further improve the essay scoring task and the whole pipeline.

7 Analyzing rubric level scores for feedback

This section analyzes the rubric level scores by human graders to develop a set of low-level features that impact the scoring task on that particular rubric. This set of low-level features could serve as feedback in an essay scoring task. For this purpose, we consider using the rubric level and holistic score in the HEW dataset to train a Random Forest regressor in

Table 2 The Rubric level model trained on HEW dataset

Rubric on which model is trained	Test set	
	Manually transcribed	OCR-transcribed
Critical Thinking, Innovative Ideas Suggestions	0.54	0.51
Focus, Clarity and Details on the topic	0.54	0.53
Importance of Fuel Conservation for India	0.43	0.40
Presentation, Cohesiveness, Language, Grammar	0.44	0.44
Total Marks	0.58	0.56

The results show average QWK of 10 runs. The model is trained only with the manually transcribed essays. The test set results are shown for both manually transcribed and OCR transcribed essays

a supervised learning setting with the set of handcrafted features explained in Section 6.2. We also analyze the impact of the subset of features and elimination of a set of features on rubric level scoring models in Sections 7.1 and 7.2, respectively. Following is a description of the settings and rubric level experiments with the handcrafted features that we are using in all the experiments in this section.

We train five individual models, one for each rubric, including the total marks or holistic score. Table 2 shows the average QWK of the test set for the model trained with normalized rubric level scores as ground truth. The input feature set is the exclusive handcrafted features detailed in Section 6.2. Again, in the same way, as in Section 6, the training set consists of manually transcribed essays while the testing set includes manually transcribed and OCR transcribed essays.

As shown in Table 2, the average QWK is close with manually transcribed, and OCR transcribed test set pipeline for all the models. The highest QWK is for the holistic score, i.e., the “Total marks” rubric. The “Critical Thinking, Innovative Ideas Suggestions” and “Focus, Clarity and Details on the topic” rubrics show a QWK of 0.54 for manually transcribed essays. These two rubrics account for 30% each in the “Total marks” rubric. The other two rubrics, “Importance of Fuel Conservation for India” and “Presentation, Cohesiveness, Language, Grammar” show closer values for the corresponding test set for manually transcribed essays. These two rubrics account for 20% each in the “Total marks” rubric. The OCR transcribed essays show slightly less QWK than the manually transcribed essays for all the rubrics in Table 2. This observation is expected as our OCR trained in Section 4 has some transcription errors, making a slight difference in scoring.

7.1 Feature level feedback with rubric analysis

In the extended experiments, we use a subset of handcrafted features and a trained rubric level model for all the rubrics mentioned earlier. In the first setting, we have used individual features and trained the model as a supervised learning task with the normalized rubric level scores as the ground truth.

Figure 8 shows the QWK of the manually transcribed test set essays on the x-axis with a model trained with only a single feature set mentioned on the y-axis of each plot. Figure 8a shows the highest QWK for the rubric “Critical Thinking, innovative ideas, Suggestion” is achieved with the Readability and Coherence features used as the only feature set for model training. For all the other rubrics, the model trained with the feature set coherence and sentence-based features shows highest test set QWK score. The sentence-based features are

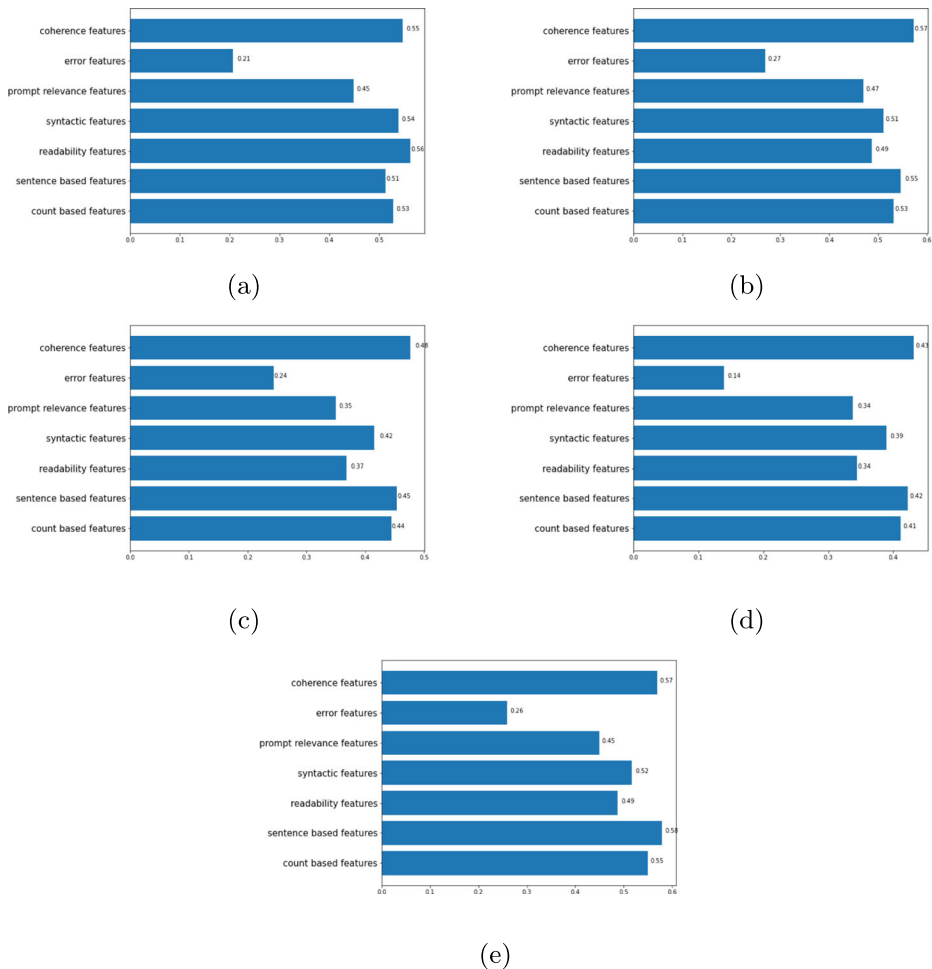


Fig. 8 QWK of the model trained with single features as marked on y-axis. Each model is trained with the rubric mentioned in the dataset: (8a) Critical Thinking, Innovative Ideas Suggestions, (8b) Focus, Clarity and Details on the topic, (8c) Importance of Fuel Conservation for India, (8c) Presentation, Cohesiveness, Language, Grammar, (8d) Total Marks, i.e. Holistic Score

designed to measure the sentences' complexity and hence can impact the clarity a sentence offers on the topic. Further, the model trained with only error features does not show a good agreement with the ground truth score on any of the rubric.

We further analyzed the impact of various combinations of features on the rubric level scores. In this setting, we exhaustively trained individual models with all possible combinations of features and chosen the model with the highest QWK on the manually transcribed essay. The set of features to train this model for a particular rubric and corresponding QWK for manually transcribed and OCR transcribed test set is shown in Table 3. In this experiment, we came up with the feature set directly impacting the agreement metric, i.e., QWK on the same test set for a given rubric. From the table, the sentence-based features appear

Table 3 The Rubric level model trained on HEW dataset

Rubric on which model is trained	Feature subset for highest QWK on Manually transcribed essays	QWK with Test Set	
		Manually Transcribed	OCR Transcribed
Critical Thinking, Innovative Ideas Suggestions	count based features sentence based features syntactic features prompt relevance features	0.58	0.50
Focus, Clarity and Details on the topic	coherence features sentence based features readability features prompt relevance features	0.59	0.57
Importance of Fuel Conservation for India	sentence based features coherence features	0.49	0.44
Presentation, Cohesiveness, Language, Grammar	count based features sentence based features readability features coherence features	0.45	0.47
Total Marks	sentence based features readability features syntactic features prompt relevance features coherence features	0.59	0.54

The subset of features that gave highest agreement, i.e., QWK with ground truth is mentioned for each rubric level model

important for all the rubrics. This feature was also among the top feature set in the experiment, where we selected individual features to train the model. The Coherence features appear to impact the scoring on all the rubrics except the “Critical Thinking, Innovative Ideas Suggestions”. In this experiment, we came up with different combinations of the feature set that collectively impact the rubric level scores.

7.2 Feature elimination

This section will show the impact of training each rubric level model by eliminating a single feature at a time from the complete set of handcrafted features mentioned in the earlier section. Figure 9 shows the QWK between the model assigned and the ground truth score for the manually transcribed test set on the x-axis. The orange bar shows the QWK with the entire set of handcrafted features, while the blue bar shows the QWK in the absence of the feature marked on the y-axis. We will refer to the results with all handcrafted features as the baseline in this section. The sub-figures are for individual rubrics, as mentioned.

In this experiment, eliminating a single feature set impacts the scoring differently for each rubric, implying the importance of the feature for a given rubric. For the rubric “Critical Thinking, Innovative Ideas Suggestions”, eliminating prompt-relevance features does

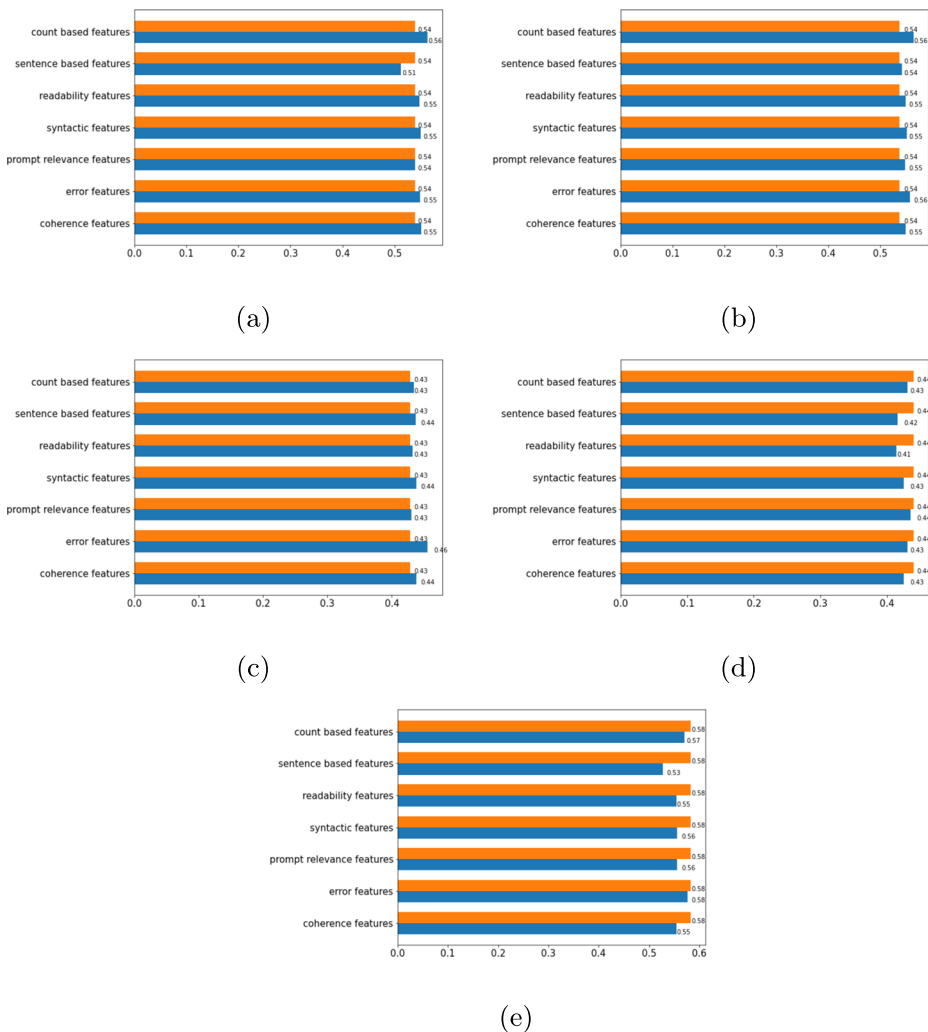


Fig. 9 QWK, of the manually transcribed test set essays, for a model trained with all features except the one marked on the y-axis (Blue bar). The orange bar shows the QWK for a model trained with all handcrafted features. Each sub-figure corresponds to model trained with the rubric mentioned in the dataset: (9a) Critical Thinking, Innovative Ideas Suggestions, (9b) Focus, Clarity and Details on the topic, (9c) Importance of Fuel Conservation for India, (9d) Presentation, Cohesiveness, Language, Grammar, (9e) Total Marks, i.e. Holistic Score

not impact the QWK score than the baseline. The elimination of sentence-based features reduces the QWK score, while eliminating other features on this rubric improves the score compared to the baseline. In this setting, the absence of a single feature helps train a better essay scoring model. In contrast, Table 3 shows these features appear essential for the same rubric, in a subset formed of a combination of different features, to train a better model.

Similar observations can be applied to other rubrics and feature combinations. The absence of error features reduces the QWK score compared to the corresponding baseline with the rubrics “Presentation, Cohesiveness, Language, Grammar” and “Total marks”. At

the same time, it increases the QWK score for all other rubrics. This observation directly signifies that human graders have consciously considered these features while scoring these two rubrics. In our experiments, the absence of the other feature on these two rubrics reduces the QWK score compared to the baseline. For the other rubrics, as shown in Fig. 9a–c, the absence of most of the individual feature shows marginally better QWK than the corresponding baseline. For the rubrics “Critical Thinking, Innovative Ideas Suggestions”, “Focus, Clarity and Details on the topic”, and “Importance of Fuel Conservation for India”, a combination of different features is important which indicated rubric level importance of these features. An elementary analysis is presented here. Further experiments can be done to measure the impact each feature brings to the QWK score on the rubric level.

8 Discussion

In this work, we have shown the results of the handwritten essay scoring on an in-the-wild dataset. The pre-trained SFR, a full-page handwriting recognition, model could not detect the start of the line and segment the text line on the HEW dataset. The major challenge with the HEW dataset lies in the variability it has. The results in Section 4 show that the SFR model, when fine-tuned on the HEW dataset, shows a CER of 0.21. This CER on the HEW dataset is much higher than the CER of 0.06 reported by [34] on the IAM dataset with an image input size of $w \times 60$. We believe the CER of our model would improve on the HEW dataset when we change the input size from $w \times 32$ to $w \times 60$.

The HEW dataset has some essays where students have made submissions with a few strike-off words, pages with watermarks, and illustrations. In the current settings, we have considered to use only the text regions and omitted the figures and illustrations in the essays for scoring. Section 5 demonstrated that including a preprocessing pipeline, to reduce or remove the artifacts like watermark and back imprints, does enhance the quality of the document visibly. However, the overall average transcription performance on the test set does not improve either on CER or WER metric by adding a standalone preprocessing pipeline. The incidental improvement of the transcription performance on the page images with high impact due to such artifacts implies to use a human-in-the-loop approach to decide on application of this pre-processing pipeline.

The essay scoring model trained from scratch for the ASAP-AES dataset shows a much better QWK than the HEW essay dataset. The ASAP-AES dataset has eight different prompts, and the QWK on each prompt had a wide variation; 0.52 on prompt8 being minimum and 0.85 on prompt1 being maximum among all prompts. The QWK of the model trained from scratch and tested on manually transcribed essays is in this range which does not necessarily say the HEW dataset is complex in terms of essays. Though the model performance improved when the ASAP-AES dataset is used for pre-training and as mixed training set for contextual embeddings as features and handcrafted features, respectively. In the future, we would experiment on a much larger set from the HEW dataset. Also, training on a larger dataset will improve the performance of the essay scoring model.

9 Conclusion

Recent research on handwriting recognition and automated essay scoring have shown tremendous progress on several publicly available datasets. In this work, we assess the

performance of handwritten essay scoring on an unconstrained dataset. The dataset is a collection of essays written by school students for a national-level essay competition. To adapt to the unconstrained nature of the dataset, we fine-tune an existing full-page handwriting recognition model. We further compare the transcriptions against the manual ground truth by the essay scoring pipeline. We consider both handcrafted and pre-trained contextual embedding features for the essays and compare them for essay scoring. In our experiments, both sets of features give closer values of agreement with human graders (measured on QWK). We also explored the impact of individual and subset of features on rubric level score. The OCR transcription errors do not affect the essay scoring task by a large margin. Simultaneously, the performance of the essay scoring model can further be improved, which needs more reliable and interpretable features. We will extend the rubric-level experiments with a larger set of essays from the in-the-wild dataset to make more insightful feedback available to students in the future.

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